

Lab 3: Architectural Justification

Virtual Technical Conference Platform

Kruthiknandan R | SRN: PES1UG24CS239

Architecture Selection

We chose Microservices Architecture for the Virtual Technical Conference Platform System.

Reason 1: Independent Scaling for Unpredictable Load

A virtual conference generates sharply uneven load across its features. Thousands of attendees may join the Live Streaming Service simultaneously during a keynote, while the Chat & Q&A Service and Payment Service see comparatively light and steady traffic. A microservices architecture allows the Live Streaming Service to be scaled independently (e.g., extra instances or CDN edge nodes) without over-provisioning the rest of the system, which would be wasteful and difficult under a monolithic or purely layered design.

Reason 2: Independent Deployment and Technology Diversity

Each capability of the platform has different technical needs: the Live Streaming Service benefits from low-latency media protocols (e.g., WebRTC/RTMP), while the Registration Service and Payment Service are better served by standard REST APIs with strong transactional guarantees. Microservices let each team build, test, and deploy its service independently and choose the technology stack best suited to it, so a bug fix in the Chat & Q&A Service can be released without redeploying or risking downtime in the Payment Service during an active conference session.

Security Advantage

Microservices support fault and security isolation: the Payment Service can be deployed in its own isolated network segment with stricter access control and PCI-relevant hardening, separate from public-facing services such as Chat & Q&A. If the Live Streaming Service or Chat Service is compromised or attacked (e.g., a DDoS attempt during a popular session), the breach is contained within that service's boundary and does not directly expose sensitive payment or user credential data held by other services.

Performance Benefit

Because each service runs and scales independently, the Live Streaming Service can be replicated across multiple servers or CDN nodes purely for video delivery, keeping stream latency low even under peak attendee load, without being slowed down by unrelated processing such as receipt generation in the Registration Service or database writes in the Chat & Q&A Service. This isolation of workloads prevents a spike in one feature (e.g., chat activity during a Q&A) from degrading the performance of another (e.g., video playback).